

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.3 TAXONOMY II: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can we identify and classify living organisms accurately? In this task you will use what you observed and learned in Lessons 1 to 8 to write your own Final Explanation. Answer each prompt in the blank space under it. Use the organisms from our school environment (maize, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat) as your evidence. Use the Claim-Evidence-Reasoning (CER) pattern where asked: make a claim, support it with evidence you observed, and give reasoning that links the evidence to the claim. Write scientific terms in English. You may talk through your ideas first in Kiswahili or your mother tongue. At the end, think about how your answers will help your group produce its Local Biodiversity Classification Guide.

Part 1: The problem with our first groupings

In Lesson 1 your group sorted the organisms from around the school (maize, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat). (a) Describe how your group sorted them and how other groups' groupings compared with yours. (b) Explain why a standard system of classification is needed so that any person anywhere can identify and study these organisms.

(a) Our group sorted the organisms by size and by whether we use them as food. Other groups sorted by colour or by habitat, so their groups were different from ours. The same organism, for example the maize plant or the snail, ended up with different neighbours in each group. No single grouping worked for everyone. (b) A standard system is needed because personal rules such as size, colour, habitat or usefulness change from person to person, so people get different answers. A standard system uses the same shared, observable characteristics for everyone, so any person anywhere who checks the same features will place an organism in the same group. This lets scientists identify, study and communicate about organisms accurately.

Part 2: Classifying plants

Use the maize plant, bean plant, fern and moss to answer. (a) State the observable features that place a plant in Bryophyta, Pteridophyta or Spermatophyta. (b) Use CER to justify the

(a) Bryophyta (moss): small size, rhizoids instead of true roots, no vascular tissue, spores in capsules, and they grow in damp places. Pteridophyta (fern): true roots, a rhizome, vascular tissue, fronds with sori on the underside, reproduce by spores, and have no seeds or flowers. Spermatophyta (maize, bean): roots, stems and leaves with vascular tissue, and they reproduce by seeds (grains and pods). Maize is a monocot and bean is a dicot. (b) Claim: The moss and the fern belong to different divisions. Evidence: Under the hand lens the moss had rhizoids, no true roots and no vascular tissue, and it had spore capsules. The fern had true roots, a rhizome and fronds with brown sori on the underside. Reasoning: These are shared features anyone can check.

placement of the moss and the fern in different groups. Your evidence must come from what you observed in Lessons 2 and 3.	The moss lacks vascular tissue and true roots, which is the Bryophyta pattern. The fern has vascular tissue and true roots but reproduces by spores, not seeds, which is the Pteridophyta pattern. So the features, not size or colour, decide the division.
---	---

Part 3: Classifying animals

Look at the earthworm, snail, mosquito, butterfly, chicken, lizard and goat. (a) Which observable body features can be used to place these animals into different major phyla? (b) The goat and the chicken are both Chordata. Explain how they can still be separated, and name the class of each. (c) Write a CER statement explaining why the mosquito is placed in Arthropoda.	(a) Annelida: soft, segmented body with no legs (earthworm). Mollusca: soft body with a shell (snail). Arthropoda: jointed legs and a hard outer covering called an exoskeleton (mosquito, butterfly). Chordata: a backbone (chicken, lizard, goat). (b) Phyla are divided into classes using more specific features such as body covering, limbs, respiration and reproduction. The goat has hair and is a mammal (Mammalia), while the chicken has feathers and is a bird (Aves). The other classes of Chordata are Pisces, Amphibia and Reptilia (the lizard is Reptilia). (c) Claim: The mosquito belongs to Arthropoda. Evidence: It has jointed legs and a hard outer exoskeleton, like the butterfly. Reasoning: Jointed legs with an exoskeleton are the defining features of Arthropoda, and the mosquito has no backbone, so it cannot be Chordata. Its placement depends on checkable features, not on its small size.
---	--

Part 4: Dichotomous keys and viruses

(a) Explain how a dichotomous key works and why different people using the same key get the same answer, while groupings by size, colour or usefulness do not. (b) Write the first two couplets of a dichotomous key that could separate the maize plant, fern and moss, using features you can check by eye or with a hand lens. (c) Viruses are not placed with plants or animals. State their characteristics and give two reasons why they are important in nature or to people in Kenya.	(a) A dichotomous key is a set of numbered couplets. Each couplet gives two contrasting, observable features, and each choice leads either to another couplet or to a name. Because every step has only two clear choices based on features anyone can check, all users follow the same path and reach the same identification. Size, colour or usefulness are opinions and vary between people, so they give different groupings. (b) 1a. Plant is small, with rhizoids at the base (seen with a hand lens) and spore capsules, and no true roots: moss (Bryophyta). 1b. Plant has true roots, stems and leaves: go to 2. 2a. Plant has fronds with brown sori on the underside and no flowers or seeds: fern (Pteridophyta). 2b. Plant has flowers and produces seeds (grains or pods): maize (Spermatophyta). (c) Viruses are non-cellular, have a protein coat and genetic material, and reproduce only inside host cells, so they differ from cellular organisms. Two reasons they are important: (1) they cause diseases that affect health and farming, such as HIV, measles, COVID-19, maize lethal necrosis disease in Kenyan maize and cassava mosaic disease; (2) they are useful, for example in making vaccines such as the measles and polio vaccines and in biotechnology.
--	--

Part 5: Final Explanation and your Local Biodiversity Classification Guide

Write a Final Explanation that answers the driving question: How can we identify and classify living organisms accurately? Use the CER pattern. Include (a) a claim, (b) evidence from at least two different lessons and organisms from our school environment, (c) reasoning that links your evidence to the claim, and (d) how your Lesson 1 groupings changed. Then say which features of your group's Local Biodiversity Classification Guide (examples, classification, features used, plant key, animal key, viruses) show this idea.	Claim: We can identify and classify living organisms accurately by using a standard system based on shared, observable characteristics, together with tools such as dichotomous keys. Evidence: In Lesson 1 groups that sorted the school organisms by size, colour or usefulness got different groupings. In Lessons 2 and 3 we saw that moss has rhizoids and no vascular tissue, the fern has true roots and sori, and maize and bean have seeds, so they fall into Bryophyta, Pteridophyta and Spermatophyta. In Lessons 5 and 6 the earthworm, snail, mosquito and chicken, lizard and goat were placed in Annelida, Mollusca, Arthropoda and Chordata, and then into classes. In Lessons 4 and 7 different groups using the same dichotomous key reached the same names. Reasoning: Shared, checkable features give everyone the same rules, so any person anywhere reaches the same result. Personal rules do not, which is why our first groupings disagreed. Viruses, being non-cellular, show that classification depends on structure and characteristics, and they are important in health, agriculture and biotechnology. Change in thinking: In Lesson 1 each group used its own rules; now we use divisions, phyla, classes and keys. In our Local Biodiversity Classification Guide, the local plant and animal examples, the characteristics listed, the plant key and animal key, and the virus note all show this idea, and anyone can use the guide to identify the same organisms and get the same answers.
--	---

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Uses specific observations from several lessons (e.g. moss rhizoids, fern sori, earthworm segmentation, key results) and links them to the school organisms.	Uses some observations from the lessons but they are general or from only one or two lessons.	Gives little or no evidence, or evidence that is not from the lessons.
Accuracy of classification ideas	Correctly names and describes divisions, phyla and classes with the features that define them, and explains viruses accurately.	Mostly correct classification with minor gaps or mix-ups in features or terms.	Has major errors or classifies by size, colour or use instead of structural features.
Claim-Evidence-Reasoning	Gives a clear claim, relevant evidence and reasoning that explains why the evidence supports the claim.	Gives a claim and evidence, but the reasoning is brief or incomplete.	Gives a claim without evidence or reasoning, or the parts are not connected.
Dichotomous key and the need for a standard system	Builds correct couplets with contrasting observable features and explains clearly why keys give the same answer for everyone.	Builds a mostly workable key and gives a partial explanation of why keys are reliable.	Key is unclear or not paired, and the need for a standard system is not explained.
Link to the driving question and Local Biodiversity Classification	Directly answers the driving question, shows how thinking changed since	Answers the driving question and connects to some parts of the guide.	Does not clearly answer the driving question or link to the guide.

Guide	Lesson 1, and connects ideas to every part of the guide.		
-------	--	--	--

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.
